



Improving drug response prediction based on two-space graph convolution

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ARTICLE INFO

Keywords:

Anti-cancer drug response

Cancer cell line

Graph convolutional neural network

Heterogeneous network

ABSTRACT

Patients with the same cancer types may present different genomic features and therefore have different drug sensitivities. Accordingly, correctly predicting patients' responses to the drugs can guide treatment decisions and improve the outcome of cancer patients. Existing computational methods leverage the graph convolution network model to aggregate features of different types of nodes in the heterogeneous network. They most fail to consider the similarity between homogeneous nodes. To this end, we propose an algorithm based on two-space graph convolutional neural networks, TSGCNN, to predict the response of anticancer drugs. TSGCNN first constructs the cell line feature space and the drug feature space and separately performs the graph convolution operation on the feature spaces to diffuse similarity information among homogeneous nodes. After that, we generate a heterogeneous network based on the known cell line and drug relationship and perform graph convolution operations on the heterogeneous network to collect the features of different types of nodes. Subsequently, the algorithm produces the final feature representations for cell lines and drugs by adding their self features, the feature space representations, and the heterogeneous space representations. Finally, we leverage the linear correlation coefficient decoder to reconstruct the cell line-drug correlation matrix for drug response prediction based on the final representations. We tested our model on the Cancer Drug Sensitivity Data (GDSC) and Cancer Cell Line Encyclopedia (CCLE) databases. The results indicate that TSGCNN shows excellent performance drug response prediction compared with other eight state-of-the-art methods.

1. Introduction

Human diseases, especially cancer, are highly heterogeneous. Patients with the same cancer types may present different genomic features and therefore have different sensitivities to drugs. Therefore, it is necessary to design a personalized medication plan to improve the effectiveness of disease treatment, which requires knowing the response profile of the various drugs. Although the drug response measured experimentally has high accuracy and reliability, it is time-consuming and costly [1]. The development of high-throughput sequencing technologies produced many genomics, transcriptomics, epigenomics and drug response data on human cell lines. These data lay the foundation for the development of computational methods that to predict drug response in different cell lines [2–4]. However, these omics data are not always available for the same samples in clinical experiments. Therefore, most current methods for predicting drug response are developed based on gene expression. At the same time, Costello et al. [5] found that gene expression is one of the effective omics data for predicting

drug response by studying the experimental results of various omics data in the NCI-DREAM challenge.

In previous studies, researchers have designed many excellent algorithms to predict the response of cancer cell lines to drugs [6,7]. We can roughly divide these methods into two categories (See Table 1). **The first category methods extract the characters of drugs and cell lines through feature engineering, then input the features of cell lines and drugs into the classifier to predict the response of drugs in the cell lines.** The difference between these methods lies in feature extraction and classifier. For cell lines, deep neural network (DNN) [8], auto-encoder (AE) [9], and convolutional neural network (CNN) are popular methods to extract cell line features from the cell-line gene expression data [8–11], somatic mutation [9], and copy number variation [9,10]. For drugs, the researchers usually extract drug features from the drug fingerprint, smile sequence, or construct a molecular graph where atoms are nodes and chemical bonds are edges [8,11]. The classifiers for cell line-drug association prediction include DNN [8,9], Deep Forest [10],

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Table 1
Review of representative drug response prediction methods.

		Cell line feature	Drug feature	Classifier	Reference
Feature engineering	DeepForest	E,C	–	DF	[10]
	MOLI	E,M,C	–	DNN	[9]
	DeepCDR	E,M,C	Molecular graph	CNN	[8]
	GraphDRP	M	Molecular graph	CNN	[11]
Heterogeneous network	HNMDRP	E	Drug–target	ID	[12]
	SRMF	E	Fingerprints	MF	[13]
	MOFGCN	E,M,C	Fingerprints	GCN	[14]
	GraphCDR	E,M,C	Molecular graph	GCN	[15]

E: expression profiles; M: mutation status; C: copy number variation; DF: deep forest; DNN: dense neural network; GCN: graph convolutional network; MF: matrix factorization; ID: information diffusion.

and CNN [11]. Most feature engineering-based methods fail to use the associations between cell lines or between drugs fully.

Another prediction method is proposed based on the heterogeneous network under the hypothesis that similar drugs have similar responses in cell lines [16–18]. These algorithms usually construct a heterogeneous network consisting of a cell line similarity network, a drug similarity network, and a cell line–drug association network. They leverage the information diffusion method, matrix factorization algorithm, and network embedding method to learn the representations of cell lines and drugs based on the heterogeneous network. Then, they use the learned representations to reconstruct the cell line–drug association matrix to predict the drug response. The information diffusion methods, such as HNMDRP [12], propagate feature information along the path of the heterogeneous network to identify the drug response. The disadvantage of this method is that it may introduce false-positive associations due to false-positive connections in the network. The matrix factorization algorithm, i.e., SRMF [13], decomposes the known cell line–drug correlation matrix to learn the characteristics of cell lines and drugs to predict drug response. At the same time, the algorithm also considers the drug similarity and the cell lines similarity as the regular constraint in matrix decomposition. The matrix factorization algorithms usually think the cell line (drug) features are a linear combination of drug (cell line) features but ignore their non-linear features. The recent popular network embedding methods, such as graph convolutional neural network (GCN) model, have drawn attention from researchers to predict drug response. These GCN-based methods, such as MOFGCN [14], and GraphCDR [15], run a graph convolutional neural network model on the known cell line–drug correlation network to learn the representations of the cell lines and the drugs. Then, they use the learned representations to reconstruct the cell line–drug association matrix to predict drug response. However, these GCN-based methods fail to aggregate information from similar homogeneous nodes.

To fully utilize the information between both homogeneous and heterogeneous nodes to predict drug response, we designed a two-space graph convolution neural network (TSGCNN) algorithm. Our algorithm first constructs the cell line feature space and the drug feature space based on cell line similarity and drug similarity. It performs the graph convolution operation in the two feature spaces to obtain cell lines and drug's feature space representations through information propagation among similar homogeneous nodes. After that, we build a heterogeneous network based on the known cell line–drug associations and run a graph convolution model to get the heterogeneous space representations for cell lines and drugs by collecting information from nodes of different types. Subsequently, the algorithm generates the final representations for cell lines and drugs by summarizing their self features, feature space representations, and heterogeneous space representations. Finally, we leverage the linear correlation coefficient decoder to reconstruct the cell line–drug correlation matrix based on the final representations. We applied our model to predict drug response based on GDSC and CCLE databases. The experimental results

show that our model outperforms the other eight state-of-the-art methods, in random zeroing, new drugs, new cell lines, and single drug experiments.

In summary, our model has the following contributions:

- We proposed a two-space graph convolutional neural network model, TSGCNN, which predicts the drug response in an end-to-end way.
- We constructed cell line feature space and drug feature space and separately perform graph convolution operations in the two feature spaces to diffuse information between similar homogeneous nodes.
- We designed a heterogeneous graph convolutional network model to gather features from the nodes of different types. Moreover, we generated final feature representations for cell lines and drugs by simultaneously considering their self features, the feature space representations, and the heterogeneous space representations.
- We have carried out a lot of experiments to verify the effectiveness of our model. The experimental results show that our algorithm TSGCNN performs well in all indicators compared with the existing eight advanced algorithms.

2. Materials

2.1. Datasets

We performed experiments on two datasets, GDSC [2] and CCLE [4], to test our model. There are two critical tables in the GDSC database: Table S4A and Table S5C. A drug is considered as being sensitive in a cell line if its IC50 value in Table S4A is above its threshold in Table S5C. Otherwise, the drug is resistant to the cell line [9,14]. Hence, we obtained 990 cancer cell lines and 265 drugs from the GDSC database, including 20,851 sensitive and 156,512 resistant samples. For the CCLE dataset, the log10(IC50) values were Z-score normalized to zero mean and unit variance across all cell lines. A cell line is defined as sensitive to a drug if the Z-score normalized log10 (IC50) value is lower than −0.8. Otherwise, the cell line is resistant to the drug. Therefore, we obtained 1696 sensitive and 8768 drug-resistant samples meeting the criteria in the CCLE dataset [14,19]. According to the literature [9], we extracted the gene expression, copy number variation and somatic mutation histological features of the corresponding cell lines from GDSC. In addition, we removed cell lines lacking any histological data and drugs sharing the same compound ID (CID) from PubChem [20]. After processing, we obtained gene expression, copy number variation and somatic mutation histological profiles for 962 cell lines, and their responses to 228 drugs.

2.2. Cell line–cell line similarity

This work measures the similarity between cell lines based on their gene expression profiles. Before calculating the cell line similarity, we

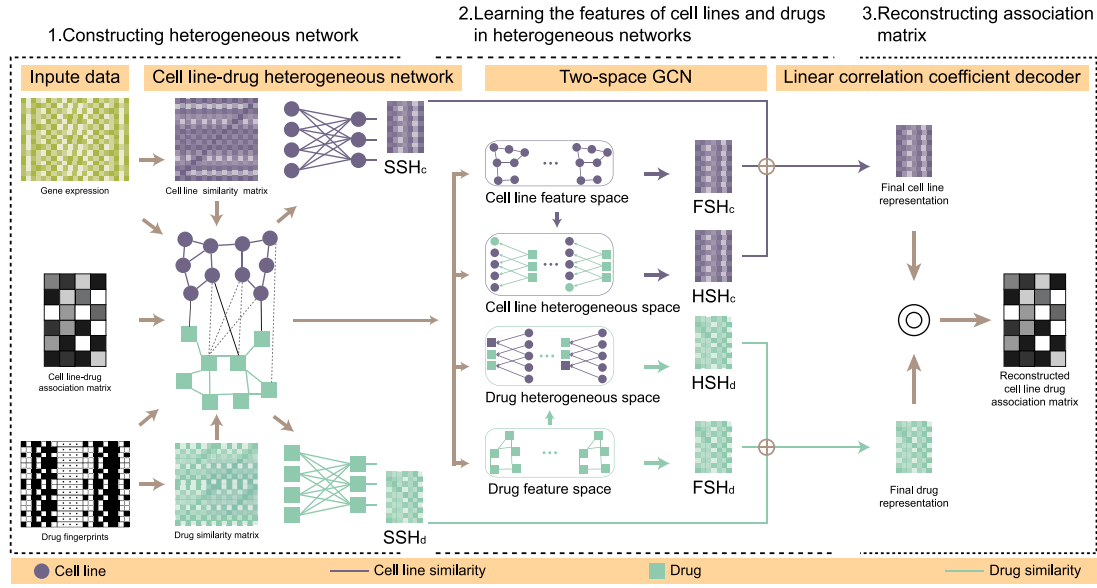


Fig. 1. Fig. 1 is the workflow of TSGCNN. TSGCNN consists of three modules: constructing a heterogeneous network, learning latent features of cell lines and drugs in heterogeneous networks, and reconstructing the cell line-drug association matrix.

perform normalization on the expression values of every gene in all cell lines.

$$\bar{x}_k = \frac{expr_k - \mu_k}{\sigma_k} \quad (1)$$

where $expr_k$ is a vector, representing the expression abundance of the k th gene in all cell lines. μ_k and σ_k represent the mean and standard deviation of $expr_k$, respectively. Based on the normalized gene expression values, the similarity between cell lines i and j is defined as follows:

$$S_{c(ij)} = e^{-\frac{\|x_i - x_j\|_2}{2e^2}} \quad (2)$$

If there are m cell lines, then $S_c \in \mathbb{R}^{m \times m}$ and x_i represents the normalized gene expression values of all genes in the i th cell line, ϵ represents the regularization constant, and we take $\epsilon = 3$ in the experiment.

2.3. Drug-drug similarity

The molecular fingerprints are commonly used as drug feature descriptors, which encode the properties of small drug molecules into binary bit strings with 920 in length. We assess the similarity of drugs i and j based on the Jaccard coefficient of their molecular fingerprints (see Eq. (3)) because the Jaccard coefficient is suitable to measure the similarity between two discrete vectors.

$$S_{d(ij)} = \frac{|d_i \cap d_j|}{|d_i \cup d_j|} \quad (3)$$

If there are n drugs, then $S_d \in \mathbb{R}^{n \times n}$, and d_i represent the molecular fingerprint features of the i th drug.

3. Two-space graph convolution

Our model TSGCNN aims to learn feature representations for cell lines and drugs and leverages the learned representations to reconstruct the cell line-drug associations indicating the drug response to cell lines. We learn the features representations from three perspectives, i.e., aggregating features from the nodes with the same or different node types and themselves initial features. For the convenience of description, let $G(c, d, e)$ describe the complicated relationship between cell lines and drugs, where c and d represent two types of nodes, cell

lines and drugs, respectively, and e represents the edges between nodes with the same or different node types. We use matrix $A \in \mathbb{R}^{(m \times n)}$ as the cell line-drug association matrix, where m denotes the number of the cell line, n represents the number of drugs. $A_{ij} = 1$ means drug j is sensitive in cell line i . Otherwise, $A_{ij} = 0$. Let matrix AC and AD be the adjacent matrix of the cell line feature space and drug feature space, which are defined based on the cell line similarity and drug similarity (see Eqs. (5) and (6)). We used cell line gene expression and drug molecular fingerprints as cell line input features and drug input features of the model, respectively. Since cell line gene expression and drug molecular fingerprint features usually have different dimensions, we performed a linear transformation of cell line normalized gene expression using the weight matrix W_c :

$$RC_i = C_i \cdot W_c \quad (4)$$

where $RC \in \mathbb{R}^{m \times f}$ denotes the node attribute of the cell line in the network, C_i is the gene expression of the i th cell line after normalization by Eq. (1). Let $RD = D_d \cdot W_d \in \mathbb{R}^{n \times f}$ be the node attribute of the drug.

3.1. Feature space graph convolutional

Considering that similar cell lines have similar responses to similar drugs, we want to learn feature representations for cell lines and drugs by referring to their similar homogeneous neighbors. Therefore, two feature spaces (AC and AD) are introduced based on top K_c most similar cell lines and K_d most similar drugs, respectively [21]:

$$AC_{ij} = \begin{cases} S_{c(ij)} & j \in K_c \\ 0 & j \notin K_c \end{cases} \quad (5)$$

$$AD_{ij} = \begin{cases} S_{d(ij)} & j \in K_d \\ 0 & j \notin K_d \end{cases} \quad (6)$$

where $j \in K_c$ represents that node j belongs to the K_c most similar homogeneous nodes of the cell line i . Let $D_{AC(1)(ii)} = \sum_i AC_{ij}$, $D_{AC(2)(ii)} = \sum_j AC_{ji}$, $L_{AC} = D_{AC(1)}^{-\frac{1}{2}} \cdot AC \cdot D_{AC(2)}^{-\frac{1}{2}}$; $D_{AD(1)(ii)} = \sum_i AD_{ij}$, $D_{AD(2)(ii)} = \sum_j AD_{ji}$, $L_{AD} = D_{AD(1)}^{-\frac{1}{2}} \cdot AD \cdot D_{AD(2)}^{-\frac{1}{2}}$. We perform a graph convolutional neural network [22] in the cell line feature space and the drug feature space (FeatureSpace, FS), respectively, to diffuse the similarity information between nodes of the same type. Then we learn feature representations for the cell lines and drugs by Eqs. (7) and (8).

$$FSH_c = \sigma(L_{AC} \cdot RC \cdot W_{FSC}) \quad (7)$$

$$FSH_d = \sigma(L_{AD} \cdot RD \cdot W_{FSD}) \quad (8)$$

where $L_{AC} \in \mathbb{R}^{(m \times m)}$ and $L_{AD} \in \mathbb{R}^{(n \times n)}$ are the normalized Laplace matrix of the feature space adjacent matrices AC and AD, respectively. W_* represents the weight parameter. $FSH_c \in \mathbb{R}^{m \times l}$ and $FSH_d \in \mathbb{R}^{n \times l}$ are the learned feature space representations of the cell lines and drugs, respectively, and σ is the activation function.

3.2. Heterogeneous space graph convolutional

To fully utilize the data of the known cell line-drug response, we build a heterogeneous network consisting of cell lines, drugs, and their known associations. Then we run graph convolutional operation on the network to diffuse features among nodes with different node types. Suppose $D_c = \sum_j A_{ij} + 1$, $D_d = \sum_j A_{ji} + 1$, $L_c = D_c^{-\frac{1}{2}} A D_c^{-\frac{1}{2}}$, $L_d = D_d^{-\frac{1}{2}} A^T D_d^{-\frac{1}{2}}$, then we use Eqs. (9) and (10) to learn feature representations for cell lines (SSH_c) and drugs (SSH_d) in the heterogeneous space (Heterogeneous Space, HS).

$$SSH_c = \sigma(L_c \cdot RD \cdot W_{TSC}) + L_c \cdot FSH_d \quad (9)$$

$$SSH_d = \sigma(L_d \cdot RD \cdot W_{TSD}) + L_d \cdot FSH_c \quad (10)$$

Here, $L_c \in \mathbb{R}^{(m \times m)}$ and $L_d \in \mathbb{R}^{(n \times n)}$ are the normalized Laplace matrix of cell line-drug association matrices A and A^T , respectively. W_* is the weight parameter, $TSH_c \in \mathbb{R}^{m \times l}$ and $TSH_d \in \mathbb{R}^{n \times l}$ represent the heterogeneous space representations of cell lines and drugs, and σ denotes the activation function.

3.3. Predicting drug response

We also calculate the self features (SSH) of cell lines and drugs by linearly transforming their original node features.

$$SSH_c = (D_c^{-1} + I_m) \cdot RC \cdot W_{c0} \quad (11)$$

$$SSH_d = (D_d^{-1} + I_n) \cdot RD \cdot W_{d0} \quad (12)$$

Here, $(D_c^{-1} + I_m)$ and $(D_d^{-1} + I_n)$ represent the normalized input features of cell lines and drugs.

After obtaining the feature space representations, heterogeneous space representations, and self-similarity features for cell lines and drugs, we sum them up to get their final representations $H_{(1-c)}$ and $H_{(1-d)}$:

$$H_{1-c} = SSH_c + SSH_c + FSH_c \quad (13)$$

$$H_{1-d} = SSH_d + SSH_d + FSH_d \quad (14)$$

We employ a linear correlation coefficient decoder [14] to reconstruct the cell line-drug association matrix to predict the drug response in cell lines. First, we use Eqs. (15) and (16) to map the representations of cell lines and drugs to low-dimensional space:

$$H_{2-c} = H_{1-c} \cdot W_{2-c} \quad (15)$$

$$H_{2-d} = H_{1-d} \cdot W_{2-d} \quad (16)$$

where $W_{(2-c)} \in \mathbb{R}^{(l \times k)}$ and $W_{(2-d)} \in \mathbb{R}^{(l \times k)}$ describe the weight parameters of the neural network of the cell line and the drug, respectively, and $k \ll l$. We leverage Equation (17) to calculate the correlation coefficient between the cell lines and drugs in the low-dimensional space:

$$Corr(h_i, h_j) = \frac{(h_i - \mu_i)(h_j - \mu_j)^T}{\sqrt{(h_i - \mu_i)(h_i - \mu_i)^T} \sqrt{(h_j - \mu_j)(h_j - \mu_j)^T}} \quad (17)$$

where $h_i \in H_{(2-c)}$ and $h_j \in H_{(2-d)}$ are the k -dimensional feature representation vector of the i th cell line and the j th drug, respectively. μ_i and μ_j denote the mean value of h_i and h_j , respectively. We utilize Equations (18) and (19) to activate the output:

$$f(h) = \frac{1}{1 + e^{-ah}} \quad (18)$$

$$g(h) = \frac{h - \min(h)}{\max(h) - \min(h)}, \quad h \in [-1, 1] \quad (19)$$

where α is the scaling parameter that controls the model gradient. The smaller the α is, the smaller the absolute value of the model gradient is, and vice versa. When $\alpha = 1$, it degenerates into a sigmoid function. Our experiment sets $\alpha = 6.9$ to make the model parameters update relatively fast in backpropagation. Finally, the reconstructed cell line-drug association matrix can be expressed as:

$$\hat{A} = g(f(Corr(H_{2-c}, H_{2-d}))) \quad (20)$$

A cross-entropy loss function is designed to optimize the model parameters.

$$\mathbb{L} = -\frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n \{M_{ij}(\beta \cdot A_{ij} \cdot \log(\hat{A}_{ij})) + (1 - A_{ij}) \cdot \log(1 - \hat{A}_{ij})\} \quad (21)$$

The matrix M indicates whether the cell line-drug associations are in the training set. $M_{ij} = 1$ if the relationship between the i th cell line and the j th drug is used for training. Otherwise, $M_{ij} = 0$. β balances the weight of the positive sample. In our experiment, we set $\beta = 1.75$.

We implement our model based on the Pytorch framework and determine the optimal hyper parameters of the model through grid search. We chose the parameter size with the highest AUC prediction result. In our experiments, we set the scale parameter $\epsilon = 3$, $K_c = 7$ and $K_d = 7$ in the feature space, the embedding space dimension $l = 224$, the decoding embedding dimension $k = 192$, the scaling parameter $\alpha = 6.9$, the learning rate as 5×10^{-4} , and the maximum number of epochs as 3000.

4. Result

4.1. Experimental design

We selected eight state-of-the-art methods for comparison. The eight methods include **MOLI** [9], **DeepForest** [10], **HNMDRP** [12], **SRMF** [13], **MOFGCN** [14], **DGCN** [23], **GraphCDR** [15], **DeepCDR** [8], and **DeepDSC** [24]. We employed the area under the receiver operating characteristic curve (AUC), average accuracy rate (AP), accuracy rate (ACC), F1 score, and Matthews correlation coefficient (MCC) to evaluate the performance of the model. Mathematically, these metrics are defined as follows.

$$ACC = \frac{TP}{TP + TN} \quad (22)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (23)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (24)$$

$$F1 \text{ Score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = \frac{2TP}{2TP + FP + FN} \quad (25)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (26)$$

Where true positive (TP), true negative (TN), false negative (FN), and false positive (FP) are four confusion matrix categories, which are calculated by comparing the predicted with the actual response values. The F1 Score is defined as the summed average of Precision and Recall. MCC will only produce a high score if the prediction yields good results in all four confusion matrix categories (TP, TN, FN and FP).

DeepDSC uses a stacked deep self-encoder to extract cell features from gene expression data and then combines them with drug chemistry features for drug response prediction. DeepCDR is a hybrid graph convolutional network based anticancer drug response(CDR) predictor integrating multi-omics profiles of cell lines and drugs. MOFGCN calculates cell line similarity and drug similarity as the cell line and drug's input features. Then it runs a graph convolutional neural network model on the known cell line-drug association network to diffuse cell line and drug similarity and reveals potential associations between cell lines and drugs. DGCN is a bipartite graph convolutional neural network algorithm for chemical network prediction problems, which performs graph convolution operations from the internal graph structure within a single compound and the external graph structure between a group of compounds, respectively. GraphCDR is one of the latest GCN-based methods that implement graph convolution operation on the known cell line-drug association network to generate representations for cell lines and drugs. It also builds a contrastive learning task based on resistant responses between cancer cell lines and drugs to enhance the model generalization ability. We employ gene expression profiles and drug fingerprints as input of GraphCDR to learn cell line features and drug features for a fair comparison. The HNMDRP predicts the cell line-drug response based on the heterogeneous network composed of the cell lines, drugs, and drug targets. It diffuses the similarity information among cell lines and drugs to fill the cell line-drug association matrix. SRMF discovers drug sensitivity through similarity regularization matrix decomposition. For a fair comparison, DeepCDR, DeepDSC, MOFGCN, GraphCDR, DGCN, HNMDRP, and SRMF only use gene expression profiles to calculate the similarity between cell lines in this work. MOLI separately leverages neural networks to learn cell line embedding based on gene expression, copy number variation, and somatic mutation profiles and then concatenates these embeddings for drug response prediction. DeepForest uses multi-information and multi-scale scanning to transform cell line gene expression and copy number variation to high-dimensional space and then uses cascade forests and feature optimization to classify anti-cancer drug responses.

We implemented the following experiments to assess the model performance from different aspects:

- Test 1: When randomly zeroing out a certain number of known cell line-drug associations, compare our model with DeepCDR, DeepDSC, MOFGCN, GraphCDR, DGCN, HNMDRP and SRMF to test their ability to recover these missing associations;
- Test 2: When randomly blinding a row or a column of the cell line-drug association matrix, compare our model with the DeepCDR, DeepDSC, MOFGCN, GraphCDR, DGCN, HNMDRP and SRMF to assess whether they can predict a new cell line or a new drug;
- Test 3: When predicting the response of a single drug, compare the performance of our algorithm and DeepCDR, DeepDSC, MOFGCN, GraphCDR, DGCN, HNMDRP, SRMF, MOLI, and DeepForest;
- Test 4: Doing ablation study on the different varieties of our model to investigate the contribution of each component in our model;
- Test 5: Doing case study of our model to see if it can detect novel drug responses in cell lines.
- Test 6: Performing regression test under randomly zeroing cross-validation in the GDSC and CCLE datasets to test whether the models' predictions show a higher correlation with the IC50.

4.2. Randomly zeroing cross-validation

Test 1 randomly and evenly partitioned the known cell line-drug associations into five parts. 1/5 positive samples and an equal random amount of negative specimens from the association matrix were selected as the test data. All the remaining positive and negative samples were used for training. We repeated the five-fold cross-validation five times and reported the average results.

Table 2 shows the experimental results of random zeroing cross-validation of all algorithms on the GDSC and CCLE databases. Compared with the non-neural network algorithms, HNMDRP, SRMF, DeepCDR and DeepDSC. MOFGCN, GraphCDR and DGCN perform significantly better. They may adopt the graph convolution technique that helps them learn suitable representations of cell lines and drugs for the prediction task. We observed that DGCN and TSGCNN achieves higher AUC, AP, Precision, Recall, F1 Score, and MCC values than MOFGCN and GraphCDR on the two datasets when using gene expression profiles to calculate the cell line similarity. **The observed improvement in the performance of DGCN and TSGCNN compared with MOFGCN and GraphCDR may be partly attributed to the high quality of cell line and drug feature representation extracted under the two-space graph convolution.**

We also compared the runtime cost of our model and all competing methods on the same computer with 4 CPU cores and 16 GB memory. As we can see from Table 3, our approach runs much faster than DeepCDR, DeepDSC and DGCN on the GDSC and CCLE datasets and takes a little more time than MOFGCN and GraphCDR using the GCN model. TSGCNN is more time-consuming than the comparison methods due to the introduction of different spatial features but is within the practical range.

4.3. New cell line or new drug experiment

In the cell line-drug association matrix, one row represents a cell line, and one column represents a drug. Test 2 cleared on row or column of the cell-line drug association matrix, which aims to verify the algorithm's ability to predict the response of a new cell line or new drug.

We only tested for cell lines or drugs with more than ten positive samples to avoid choosing drugs with too general or too special responses [19]. Therefore, we obtained 658 out of 962 cell lines and 227 out of 228 drugs from the GDSC dataset, and 26 out of 436 cell lines and 20 out of 24 drugs from the CCLE database for experiments.

Table 4 reports the comparison results of the seven algorithms on GDSC and CCLE datasets. **TSGCNN performs best in most cases** on the GDSC and CCLE datasets. HNMDRP fails to work in the new row experiment normally. The reason is that HNMDRP only uses similar drugs and similar drug targets to infer potential cell line-drug associations and ignores the similarity between cell lines. In contrast, TSGCNN is more robust. When all associations between cell lines and drugs are unknown, the information provided by the heterogeneity space is marginally, and TSGCNN also shows better performance due to self-similarity and feature space similarity. The SRMF algorithm also performs poorly in this experiment, where the AUC only reaches 0.5807 and 0.6683 in the new cell line and new drug experiments on the GDSC database. The reason is that after clearing one row or one column of the cell line-drug association matrix, the SRMF algorithm only uses self-similarity to constrain latent features, which reduces the fitting ability of the model. DeepCDR, DeepDSC, MOFGCN, GraphCDR and DGCN, four relatively new algorithms for predicting drug response, show good performance in this experiment, but our algorithm achieves better performance in most evaluation metrics **except its new drug prediction performance in CCLE dataset is slightly lower than GraphCDR. However, our model always controls the best performance on the overall performance.**

4.4. Predict a single drug response

MOLI and DeepForest are designed to predict whether a drug or several drugs are sensitive in cell lines. In test 3, we compared the performance of our algorithm with MOFGCN, GraphCDR, DGCN, DeepDSC, DeepCDR, HNMDRP, SRMF, MOLI, and DeepForest when predicting one drug response. Like test 2, only the drugs with more than ten positive samples are considered. The test set is generated by randomly

Table 2
Comparison of every method under randomly zeroing cross-validation on two datasets.

Dataset	Algorithm	AUC	AP	ACC	F1_Score	MCC
GDSC	HNMDRP	$0.7258 \pm 3 \times 10^{-5}$	$0.7198 \pm 4 \times 10^{-5}$	$0.6302 \pm 1 \times 10^{-4}$	$0.7008 \pm 1 \times 10^{-5}$	$0.3281 \pm 7 \times 10^{-4}$
	SRMF	$0.6563 \pm 2 \times 10^{-4}$	$0.6605 \pm 5 \times 10^{-5}$	$0.5587 \pm 6 \times 10^{-4}$	$0.6731 \pm 3 \times 10^{-5}$	$0.4182 \pm 4 \times 10^{-4}$
	DeepCDR	$0.7849 \pm 5 \times 10^{-5}$	$0.7827 \pm 6 \times 10^{-5}$	$0.6886 \pm 1 \times 10^{-4}$	$0.7341 \pm 1 \times 10^{-5}$	$0.4020 \pm 1 \times 10^{-4}$
	DeepDSC	$0.8118 \pm 4 \times 10^{-4}$	$0.8311 \pm 1 \times 10^{-4}$	$0.7271 \pm 2 \times 10^{-4}$	$0.7467 \pm 2 \times 10^{-4}$	$0.4603 \pm 9 \times 10^{-4}$
	MOFGCN	$0.8684 \pm 7 \times 10^{-6}$	$0.8730 \pm 1 \times 10^{-5}$	$0.7788 \pm 3 \times 10^{-5}$	$0.7921 \pm 9 \times 10^{-6}$	$0.5627 \pm 9 \times 10^{-5}$
	GraphCDR	$0.8136 \pm 1 \times 10^{-2}$	$0.8193 \pm 7 \times 10^{-3}$	$0.7116 \pm 6 \times 10^{-3}$	$0.7483 \pm 6 \times 10^{-3}$	$0.4432 \pm 1 \times 10^{-3}$
	DGCN	$0.8518 \pm 1 \times 10^{-5}$	$0.8566 \pm 1 \times 10^{-5}$	$0.7628 \pm 3 \times 10^{-5}$	$0.7796 \pm 2 \times 10^{-5}$	$0.5326 \pm 9 \times 10^{-3}$
	TSGCNN	$0.8722 \pm 1 \times 10^{-5}$	$0.8761 \pm 1 \times 10^{-5}$	$0.7834 \pm 4 \times 10^{-5}$	$0.7960 \pm 1 \times 10^{-5}$	$0.5716 \pm 1 \times 10^{-4}$
CCLE	HNMDRP	$0.7104 \pm 1 \times 10^{-4}$	$0.6956 \pm 2 \times 10^{-4}$	$0.6328 \pm 4 \times 10^{-4}$	$0.7153 \pm 4 \times 10^{-5}$	$0.3281 \pm 7 \times 10^{-4}$
	SRMF	$0.7669 \pm 4 \times 10^{-5}$	$0.7418 \pm 2 \times 10^{-5}$	$0.6792 \pm 3 \times 10^{-5}$	$0.7441 \pm 5 \times 10^{-5}$	$0.4182 \pm 4 \times 10^{-4}$
	DeepCDR	$0.8289 \pm 1 \times 10^{-4}$	$0.8185 \pm 2 \times 10^{-4}$	$0.7567 \pm 2 \times 10^{-4}$	$0.7771 \pm 1 \times 10^{-4}$	$0.5243 \pm 7 \times 10^{-4}$
	DeepDSC	$0.8594 \pm 1 \times 10^{-4}$	$0.8607 \pm 1 \times 10^{-4}$	$0.7808 \pm 3 \times 10^{-4}$	$0.7936 \pm 1 \times 10^{-4}$	$0.5685 \pm 1 \times 10^{-3}$
	MOFGCN	$0.8608 \pm 1 \times 10^{-4}$	$0.8589 \pm 1 \times 10^{-4}$	$0.7752 \pm 3 \times 10^{-4}$	$0.7910 \pm 1 \times 10^{-4}$	$0.5595 \pm 1 \times 10^{-3}$
	GraphCDR	$0.8474 \pm 2 \times 10^{-4}$	$0.8495 \pm 2 \times 10^{-4}$	$0.7702 \pm 5 \times 10^{-4}$	$0.7871 \pm 2 \times 10^{-4}$	$0.5494 \pm 1 \times 10^{-3}$
	DGCN	$0.8391 \pm 1 \times 10^{-2}$	$0.8434 \pm 7 \times 10^{-3}$	$0.7716 \pm 4 \times 10^{-4}$	$0.7829 \pm 2 \times 10^{-4}$	$0.5481 \pm 1 \times 10^{-3}$
	TSGCNN	$0.8650 \pm 1 \times 10^{-4}$	$0.8644 \pm 2 \times 10^{-4}$	$0.7859 \pm 2 \times 10^{-4}$	$0.7981 \pm 1 \times 10^{-4}$	$0.5786 \pm 8 \times 10^{-4}$

Table 3
Comparison of running time of every method (s).

Dataset	HNMDRP	SRMF	DeepCDR	DeepDSC	MOFGCN	GraphCDR	DGCN	TSGCNN
GDSC	0.04	11.87	216.71	153.28	5.04	13.83	62.14	37.77
CCLE	0.01	2.56	11.60	11.19	4.18	3.48	20.44	8.73

Table 4
Comparison of every method predicting response for a new cell line or a new drug on two datasets.

Dataset	Algorithm	AUC	AP	ACC	F1_Score	MCC
GDSC	new cell line	HNMDRP	–	–	–	–
		SRMF	$0.5807 \pm 1 \times 10^{-2}$	$0.6153 \pm 1 \times 10^{-2}$	$0.5859 \pm 7 \times 10^{-3}$	$0.6990 \pm 1 \times 10^{-3}$
		DeepCDR	$0.7526 \pm 8 \times 10^{-3}$	$0.7664 \pm 8 \times 10^{-3}$	$0.7158 \pm 9 \times 10^{-3}$	$0.7621 \pm 3 \times 10^{-3}$
		DeepDSC	$0.7831 \pm 8 \times 10^{-3}$	$0.7994 \pm 8 \times 10^{-3}$	$0.7471 \pm 9 \times 10^{-3}$	$0.7802 \pm 4 \times 10^{-3}$
		MOFGCN	$0.7190 \pm 5 \times 10^{-3}$	$0.7366 \pm 5 \times 10^{-3}$	$0.6876 \pm 5 \times 10^{-3}$	$0.7472 \pm 1 \times 10^{-3}$
		GraphCDR	$0.7122 \pm 9 \times 10^{-3}$	$0.7061 \pm 9 \times 10^{-3}$	$0.6874 \pm 7 \times 10^{-3}$	$0.7488 \pm 2 \times 10^{-3}$
		DGCN	$0.7982 \pm 8 \times 10^{-3}$	$0.8088 \pm 8 \times 10^{-3}$	$0.7566 \pm 9 \times 10^{-3}$	$0.7900 \pm 4 \times 10^{-3}$
		TSGCNN	$0.8113 \pm 6 \times 10^{-3}$	$0.8184 \pm 8 \times 10^{-3}$	$0.7698 \pm 8 \times 10^{-3}$	$0.7984 \pm 4 \times 10^{-3}$
	new drug	HNMDRP	$0.6951 \pm 1 \times 10^{-2}$	$0.6935 \pm 1 \times 10^{-2}$	$0.6496 \pm 1 \times 10^{-2}$	$0.7290 \pm 2 \times 10^{-3}$
		SRMF	$0.6683 \pm 6 \times 10^{-3}$	$0.6757 \pm 6 \times 10^{-3}$	$0.6033 \pm 6 \times 10^{-3}$	$0.7033 \pm 1 \times 10^{-3}$
		DeepCDR	$0.7605 \pm 9 \times 10^{-3}$	$0.7565 \pm 1 \times 10^{-2}$	$0.7031 \pm 8 \times 10^{-3}$	$0.7532 \pm 2 \times 10^{-3}$
		DeepDSC	$0.7472 \pm 1 \times 10^{-2}$	$0.7514 \pm 1 \times 10^{-2}$	$0.6942 \pm 9 \times 10^{-3}$	$0.7455 \pm 2 \times 10^{-3}$
		MOFGCN	$0.7601 \pm 7 \times 10^{-3}$	$0.7558 \pm 8 \times 10^{-3}$	$0.7006 \pm 6 \times 10^{-3}$	$0.7513 \pm 2 \times 10^{-3}$
		GraphCDR	$0.7614 \pm 8 \times 10^{-3}$	$0.7501 \pm 9 \times 10^{-3}$	$0.7031 \pm 6 \times 10^{-3}$	$0.7533 \pm 2 \times 10^{-3}$
		DGCN	$0.7259 \pm 6 \times 10^{-3}$	$0.7227 \pm 7 \times 10^{-3}$	$0.6673 \pm 8 \times 10^{-3}$	$0.7290 \pm 2 \times 10^{-3}$
		TSGCNN	$0.7758 \pm 5 \times 10^{-3}$	$0.7703 \pm 6 \times 10^{-3}$	$0.7200 \pm 6 \times 10^{-3}$	$0.7578 \pm 2 \times 10^{-3}$
CCLE	new cell line	HNMDRP	–	–	–	–
		SRMF	$0.6138 \pm 8 \times 10^{-3}$	$0.6187 \pm 1 \times 10^{-2}$	$0.7055 \pm 3 \times 10^{-3}$	$0.7746 \pm 1 \times 10^{-3}$
		DeepCDR	$0.8830 \pm 6 \times 10^{-3}$	$0.8913 \pm 6 \times 10^{-3}$	$0.8527 \pm 6 \times 10^{-3}$	$0.8722 \pm 4 \times 10^{-3}$
		DeepDSC	$0.8935 \pm 4 \times 10^{-3}$	$0.9073 \pm 4 \times 10^{-3}$	$0.8602 \pm 5 \times 10^{-3}$	$0.8743 \pm 3 \times 10^{-3}$
		MOFGCN	$0.8108 \pm 8 \times 10^{-3}$	$0.8137 \pm 8 \times 10^{-3}$	$0.8060 \pm 5 \times 10^{-3}$	$0.8381 \pm 3 \times 10^{-3}$
		GraphCDR	$0.7613 \pm 1 \times 10^{-2}$	$0.7694 \pm 1 \times 10^{-2}$	$0.7441 \pm 1 \times 10^{-2}$	$0.7971 \pm 4 \times 10^{-3}$
		DGCN	$0.8657 \pm 5 \times 10^{-3}$	$0.8659 \pm 8 \times 10^{-3}$	$0.8406 \pm 4 \times 10^{-3}$	$0.8587 \pm 3 \times 10^{-3}$
		TSGCNN	$0.8997 \pm 9 \times 10^{-3}$	$0.9110 \pm 4 \times 10^{-3}$	$0.8608 \pm 5 \times 10^{-3}$	$0.8789 \pm 3 \times 10^{-3}$
	new drug	HNMDRP	$0.6947 \pm 6 \times 10^{-3}$	$0.6871 \pm 6 \times 10^{-3}$	$0.6498 \pm 4 \times 10^{-3}$	$0.6713 \pm 4 \times 10^{-5}$
		SRMF	$0.4873 \pm 9 \times 10^{-3}$	$0.5288 \pm 5 \times 10^{-3}$	$0.5204 \pm 1 \times 10^{-3}$	$0.6713 \pm 4 \times 10^{-5}$
		DeepCDR	$0.7389 \pm 4 \times 10^{-3}$	$0.7300 \pm 5 \times 10^{-3}$	$0.6808 \pm 4 \times 10^{-3}$	$0.7385 \pm 1 \times 10^{-3}$
		DeepDSC	$0.7315 \pm 4 \times 10^{-3}$	$0.7295 \pm 6 \times 10^{-3}$	$0.6756 \pm 5 \times 10^{-3}$	$0.7325 \pm 1 \times 10^{-3}$
		MOFGCN	$0.7215 \pm 2 \times 10^{-3}$	$0.7113 \pm 2 \times 10^{-3}$	$0.6661 \pm 1 \times 10^{-3}$	$0.7307 \pm 5 \times 10^{-3}$
		GraphCDR	$0.7506 \pm 4 \times 10^{-3}$	$0.7280 \pm 5 \times 10^{-3}$	$0.6900 \pm 5 \times 10^{-3}$	$0.7440 \pm 1 \times 10^{-3}$
		DGCN	$0.6375 \pm 3 \times 10^{-3}$	$0.6180 \pm 3 \times 10^{-3}$	$0.5980 \pm 4 \times 10^{-3}$	$0.7002 \pm 7 \times 10^{-4}$
		TSGCNN	$0.7411 \pm 3 \times 10^{-3}$	$0.7218 \pm 3 \times 10^{-3}$	$0.6976 \pm 3 \times 10^{-3}$	$0.7407 \pm 1 \times 10^{-3}$

selecting 1/5 positive samples and an equal number of negative samples from a testing drug. As for the training set, MOLI and DeepForest use the remaining 4/5 positive specimens and all remaining negative samples of the drug. For MOFGCN, GraphCDR, DGCN, DeepDSC, DeepCDR, HNMDRP, SRMF and our model, the training data consists of all remaining positive and negative samples, excluding the data of the test set. We repeated the five-fold cross-validation experiment five times. Table 4 lists the mean and variance of the AUC, AP, ACC, F1 Score, and MCC of all methods over all tested drugs on the GDSC and CCLE databases.

As Table 5 shown, we notice that the algorithms that consider both cell line and drug features, such as TSGCNN, HNMDRP, SRMF, DeepCDR, DeepDSC, GraphCDR MOFGCN and DGCN, perform better than algorithms that only consider cell line features, such as the DeepForest and MOLI. Compared with MOFGCN, GraphCDR, DGCN, HNMDRP, and SRMF, our algorithm controls the best single-drug response prediction performance. The AUC and ACC of our proposed algorithm TSGCNN reached 0.9156 and 0.8833 in the GDSC database, and reached 0.8823 and 0.8471 in the CCLE database.

Table 5
Comparison of every method for predicting a single drug response on two datasets.

Dataset	Algorithm	AUC	AP	ACC	F1_Score	MCC
GDSC	DeepForest	$0.6214 \pm 1 \times 10^{-2}$	$0.6608 \pm 7 \times 10^{-3}$	$0.6403 \pm 5 \times 10^{-3}$	$0.7025 \pm 1 \times 10^{-3}$	$0.3280 \pm 1 \times 10^{-2}$
	MOLI	$0.6841 \pm 8 \times 10^{-3}$	$0.7167 \pm 6 \times 10^{-3}$	$0.6817 \pm 4 \times 10^{-3}$	$0.7427 \pm 1 \times 10^{-3}$	$0.4072 \pm 1 \times 10^{-2}$
	SRMF	$0.7271 \pm 2 \times 10^{-2}$	$0.7484 \pm 1 \times 10^{-2}$	$0.7229 \pm 1 \times 10^{-2}$	$0.7749 \pm 4 \times 10^{-3}$	$0.4877 \pm 4 \times 10^{-2}$
	HNMDRP	$0.7411 \pm 9 \times 10^{-3}$	$0.7570 \pm 9 \times 10^{-3}$	$0.7268 \pm 6 \times 10^{-3}$	$0.7740 \pm 1 \times 10^{-3}$	$0.4954 \pm 2 \times 10^{-2}$
	DeepCDR	$0.8009 \pm 1 \times 10^{-2}$	$0.8091 \pm 1 \times 10^{-2}$	$0.7748 \pm 1 \times 10^{-2}$	$0.8063 \pm 5 \times 10^{-3}$	$0.5812 \pm 3 \times 10^{-2}$
	DeepDSC	$0.8174 \pm 1 \times 10^{-2}$	$0.8327 \pm 1 \times 10^{-3}$	$0.7901 \pm 1 \times 10^{-2}$	$0.8154 \pm 5 \times 10^{-3}$	$0.6073 \pm 3 \times 10^{-2}$
	MOFGCN	$0.8785 \pm 3 \times 10^{-5}$	$0.8866 \pm 3 \times 10^{-3}$	$0.8446 \pm 3 \times 10^{-3}$	$0.8576 \pm 2 \times 10^{-3}$	$0.7069 \pm 1 \times 10^{-2}$
	GraphCDR	$0.8076 \pm 6 \times 10^{-3}$	$0.8144 \pm 6 \times 10^{-3}$	$0.7819 \pm 5 \times 10^{-3}$	$0.8107 \pm 2 \times 10^{-3}$	$0.5946 \pm 1 \times 10^{-2}$
	DGCN	$0.9019 \pm 6 \times 10^{-3}$	$0.9061 \pm 6 \times 10^{-3}$	$0.8689 \pm 7 \times 10^{-3}$	$0.8782 \pm 5 \times 10^{-3}$	$0.7520 \pm 1 \times 10^{-2}$
	TSGCNN	$0.9156 \pm 5 \times 10^{-3}$	$0.9182 \pm 5 \times 10^{-3}$	$0.8833 \pm 6 \times 10^{-3}$	$0.8906 \pm 5 \times 10^{-3}$	$0.7779 \pm 2 \times 10^{-2}$
CCLE	DeepForest	$0.5924 \pm 4 \times 10^{-3}$	$0.6387 \pm 5 \times 10^{-3}$	$0.6183 \pm 3 \times 10^{-3}$	$0.7122 \pm 9 \times 10^{-4}$	$0.2911 \pm 1 \times 10^{-2}$
	MOLI	$0.6899 \pm 5 \times 10^{-3}$	$0.7227 \pm 6 \times 10^{-3}$	$0.6830 \pm 4 \times 10^{-3}$	$0.7430 \pm 1 \times 10^{-3}$	$0.4113 \pm 1 \times 10^{-2}$
	SRMF	$0.7403 \pm 1 \times 10^{-2}$	$0.7524 \pm 7 \times 10^{-3}$	$0.7331 \pm 5 \times 10^{-3}$	$0.7786 \pm 2 \times 10^{-3}$	$0.5112 \pm 1 \times 10^{-2}$
	HNMDRP	$0.7202 \pm 2 \times 10^{-3}$	$0.7348 \pm 2 \times 10^{-3}$	$0.7105 \pm 2 \times 10^{-3}$	$0.7612 \pm 1 \times 10^{-3}$	$0.4702 \pm 9 \times 10^{-2}$
	DeepCDR	$0.8322 \pm 8 \times 10^{-3}$	$0.8400 \pm 8 \times 10^{-3}$	$0.8047 \pm 8 \times 10^{-3}$	$0.8246 \pm 5 \times 10^{-3}$	$0.6337 \pm 2 \times 10^{-2}$
	DeepDSC	$0.8497 \pm 9 \times 10^{-3}$	$0.8649 \pm 7 \times 10^{-3}$	$0.8239 \pm 8 \times 10^{-3}$	$0.8373 \pm 5 \times 10^{-3}$	$0.6660 \pm 3 \times 10^{-2}$
	MOFGCN	$0.8442 \pm 3 \times 10^{-3}$	$0.8563 \pm 2 \times 10^{-3}$	$0.8191 \pm 2 \times 10^{-3}$	$0.8344 \pm 1 \times 10^{-3}$	$0.6583 \pm 9 \times 10^{-3}$
	GraphCDR	$0.8438 \pm 2 \times 10^{-3}$	$0.8520 \pm 3 \times 10^{-3}$	$0.8160 \pm 2 \times 10^{-3}$	$0.8317 \pm 1 \times 10^{-3}$	$0.6526 \pm 8 \times 10^{-2}$
	DGCN	$0.8713 \pm 6 \times 10^{-3}$	$0.8779 \pm 6 \times 10^{-3}$	$0.8431 \pm 6 \times 10^{-3}$	$0.8526 \pm 4 \times 10^{-3}$	$0.7018 \pm 2 \times 10^{-2}$
	TSGCNN	$0.8823 \pm 5 \times 10^{-3}$	$0.8908 \pm 5 \times 10^{-3}$	$0.8471 \pm 6 \times 10^{-3}$	$0.8585 \pm 4 \times 10^{-3}$	$0.7108 \pm 2 \times 10^{-2}$

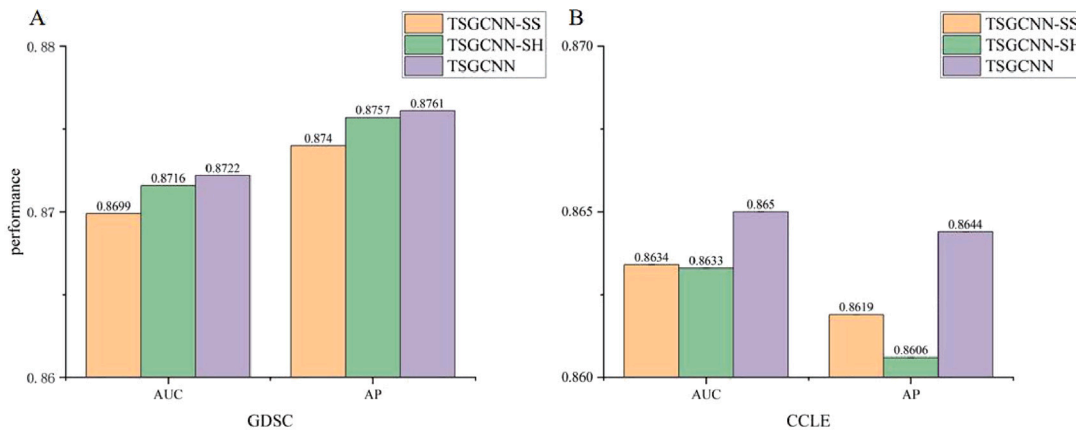


Fig. 2. Ablation study on GDSC and CCLE datasets.(A) and (B) are the AUC and AP values under GDSC and CCLE datasets, respectively.

4.5. Ablation study

Since the representations learned by our proposed algorithm are composed of three parts: feature space representations, heterogeneous space representations, and self-features. To verify the role of each component in the model, we conducted an ablation experiment in randomly zeroing cross-validation. TSGCNN-SS represents that we only use self-features to reconstruct the cell line-drug association matrix. In other words, $H_{1-c} = SSH_c$ in Eq. (13) and H_{1-d} in Eq. (14). TSGCNN-SH denotes that our method adds self features with heterogeneous space representations to reconstruct the cell line-drug association matrix. In other words, $H_{1-c} = SSH_c + HSH_c$ in Eq. (13) and $H_{1-d} = SSH_d + HSH_d$ in Eq. (14). From Fig. 2, we notice that only using the self feature to reconstruct the cell line-drug association matrix produces relatively poor results. The introduction of the heterogeneous space representation can improve the performance of our model. Moreover, our model controls the highest score by combining the three types of features.

4.6. Case study

A previous study found that 20% of the observed data had missing responses [15]. To further evaluate whether our model can detect novel drug responses in cell lines, we trained the model with all known drug responses in cell lines in the GDSC dataset and then predicted unknown responses in the GDSC data. We focused on two

clinically approved drugs, dasatinib and GSK690693. Table 6 shows the top 10 sensitive cell lines predicted by our model for dasatinib and GSK690693. We point out two cell lines RXF393 and Hey exposed to Dasatinib, which have been reported previously in the literature. Le et al. reported that dasatinib caused a small amount of apoptosis and a large amount of autophagy to inhibit cell growth in Hey ovarian cancer cells [25]. Eustace et al. reported that HT-144 was moderately sensitive to dasatinib [26]. For the drug GSK690693, three cell lines NALM-6, REH and KARPAS-231 were sensitive. Levy et al. studied the response of pre-B cell NALM-6, pre-B cell REH and B cell KARPAS-231 to the pan-akt kinase inhibitor GSK690693 [27]. According to their study, GSK690693 effectively inhibited the proliferation of NALM-6, REH and KARPAS-231.

4.7. Regression test

To verify whether the output of Eq. (17) is correlated with IC50, we added IC50 regression tests. We modified the output of the model to Eq. (17) to estimate the IC50 value of the drug and evaluated it using three regressors: Pearson correlation (PCC), Spearman correlation (SCC), and root mean square error (RMSE). We performed regression tests under cross-validation of randomized nulled values in the GDSC and CCLE datasets. Fig. 3 reports that the predictions of our model show a higher correlation with the IC50 than other competing methods.

Table 6
Top 10 predicted sensitive cell lines for Dasatinib and GSK690693.

Drug	Rank	Cell line	PMID	Drug	Rank	Cell line	PMID
Dasatinib	1	KS-1	N/A	GSK690693	1	NALM-6	19064730
	2	RXF393	21632985		2	ES7	N/A
	3	WM793B	N/A		3	REH	19064730
	4	SW982	N/A		4	HC-1	N/A
	5	OS-RC-2	N/A		5	KYAE-1	N/A
	6	WM793B	N/A		6	CESS	N/A
	7	NCI-H2369	N/A		7	RPMI-6666	N/A
	8	A204	N/A		8	KARPAS-231	19064730
	9	Hey	20629079		9	QIMR-WIL	N/A
	10	ACHN	N/A		10	SU-DHL-5	N/A

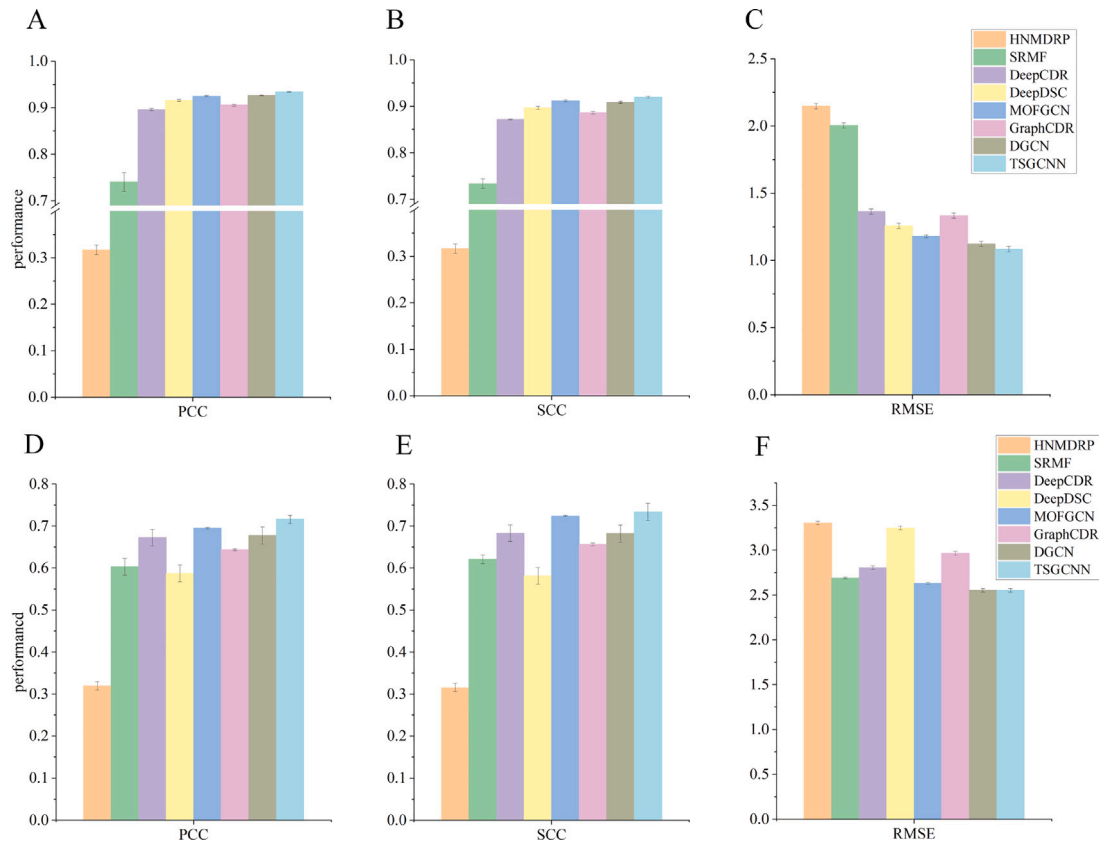


Fig. 3. The results of the randomly zeroing values cross-validation on GDSC and CCLE datasets based on regression test. (A)–(C) are Pearson correlation (PCC), Spearman correlation (SCC), and root mean square error (RMSE) on GDSC dataset, respectively. (D)–(F) are Pearson correlation (PCC), Spearman correlation (SCC), and root mean squared error (RMSE) on the CCLE dataset, respectively.

5. Conclusion

This paper proposes an algorithm based on two-space graph convolution to predict the drug response in cancer cell lines. Our algorithm first constructs the cell line feature space and the drug feature space, respectively, and performs the graph convolution operation in two feature spaces to diffuse similarity information between similar homogeneous nodes. After that, we build a heterogeneous network based on known cell line-drug associations and run a graph convolution network to collect feature information from heterogeneous nodes. Subsequently, the algorithm separately generates feature representations for cell lines and drugs by summarizing their self features and those aggregated from homogeneous or heterogeneous nodes. Finally, we employ the linear correlation coefficient decoder to reconstruct the cell line-drug correlation matrix to predict the drug response. We designed extensive experiments to evaluate the performance of our model. By analyzing the experimental results, we found that:

- Compared with the non-graph neural network models, such as HNMDRP and SRMF, our algorithm, TSGCNN, adopts graph neural network induction matrix reconstruction leading to better performance. For example, in randomly zeroing cross-validation, the AUC values of TSGNN are 0.22 and 0.15 higher than SRMF and HNMDRP on the GDSC dataset, 0.1 and 0.15 higher than SRMF and HNMDRP on the CCLE dataset. Compared with MOLI and DeepForest extracting cell line features for drug response prediction, our algorithm using both cell line and drug features shows outstanding performance in all experiments. Our model achieves 0.2 and 0.3 improvement to MOLI and Deepforest in predicting a single drug response on the two datasets.
- Compared with the previous GCN-based algorithm, i.e., MOFGCN, GraphCDR, DGCN, our model considers more information based on a two-space graph convolutional network and achieves the best performance in almost all experiments. For example, in randomly zeroing cross-validation, our model has 0.01 higher AUC values than the two methods on the GDSC and CCLE datasets.

Our future work will improve the algorithm by introducing more biological information. Inspired by the literature [28–31], we will consider using multi-task learning combined with tissue type and drug class information of cell lines to predict drug response.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported in part by the National Natural Science Foundation of China under grant No. 61972185, Natural Science Foundation of Yunnan Province of China (2019FA024), Yunnan Ten Thousand Talents Plan young.

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